SQL Lab Record

Labs 4 - 6: Functions, Operators, Special Operators and Set Operations

LAB 4 | SQL Functions

Conversion Functions - Setup

```
CREATE TABLE employees (
    employee_id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    email VARCHAR(100) UNIQUE,
    hire_date DATE,
    department_id INT
);

INSERT INTO employees VALUES
```

```
  (101, 'Arjun', 'Verma', 'arjun.v@company.com', '2020-05-10', 1),
  (102, 'Bhavna', 'Singh', 'bhavna.s@company.com', '2019-03-15', 2),
  (103, 'Chirag', 'Desai', 'chirag.d@company.com', '2021-07-20', 3),
  (104, 'Esha', 'Kapoor', 'esha.k@company.com', '2018-11-01', 4),
  (105, 'Farhan', 'Ali', 'farhan.a@company.com', '2022-01-12', 2),
  (106, 'Gauri', 'Menon', 'gauri.m@company.com', '2020-09-23', 3),
  (107, 'Harsh', 'Bansal', 'harsh.b@company.com', '2021-04-18', 1),
  (108, 'Ishita', 'Das', 'ishita.d@company.com', '2019-08-30', 4),
  (109, 'Jatin', 'Malhotra', 'jatin.m@company.com', '2022-11-05', 3),
  (110, 'Lavanya', 'Pillai', 'lavanya.p@company.com', '2023-02-10', 2);
```

QUERY

```
SELECT * FROM employees;
```

OUTPUT

employee_id	first_name	last_name	email	hire_date	department_id
101	Arjun	Verma	arjun.v@company.com	2020-05-10	1
102	Bhavna	Singh	bhavna.s@company.com	2019-03-15	2
103	Chirag	Desai	chirag.d@company.com	2021-07-20	3
104	Esha	Kapoor	esha.k@company.com	2018-11-01	4
105	Farhan	Ali	farhan.a@company.com	2022-01-12	2
106	Gauri	Menon	gauri.m@company.com	2020-09-23	3
107	Harsh	Bansal	harsh.b@company.com	2021-04-18	1
108	Ishita	Das	ishita.d@company.com	2019-08-30	4
109	Jatin	Malhotra	jatin.m@company.com	2022-11-05	3
110	Lavanya	Pillai	lavanya.p@company.com	2023-02-10	2

Implicit Data Type Conversion

QUERY

```
SELECT employee_id, first_name, department_id
FROM employees
WHERE department_id > 2;
```

OUTPUT

employee_id	first_name	department_id
103	Chirag	3
104	Esha	4
106	Gauri	3
108	Ishita	4
109	Jatin	3

Explicit Data Type Conversion

QUERY

```
SELECT employee_id, first_name,
       CAST(hire_date AS CHAR) AS hire_date_text
FROM employees;
```

OUTPUT

employee_id	first_name	hire_date_text
101	Arjun	2020-05-10
102	Bhavna	2019-03-15
103	Chirag	2021-07-20
104	Esha	2018-11-01
105	Farhan	2022-01-12
106	Gauri	2020-09-23
107	Harsh	2021-04-18
108	Ishita	2019-08-30
109	Jatin	2022-11-05
110	Lavanya	2023-02-10

Date Functions

QUERY

```sql
SELECT
    first_name,
    last_name,
    hire_date,
    CURDATE() AS today,
    YEAR(hire_date) AS year,
    MONTH(hire_date) AS month,
    DAY(hire_date) AS day,
    YEAR(CURDATE()) - YEAR(hire_date) AS approximate_years
FROM employees;
```

OUTPUT

first_name	last_name	hire_date	today	year	month	day	approximate_years
Arjun	Verma	2020-05-10	2026-10-02	2020	5	10	6
Bhavna	Singh	2019-03-15	2026-10-02	2019	3	15	7
Chirag	Desai	2021-07-20	2026-10-02	2021	7	20	5
Esha	Kapoor	2018-11-01	2026-10-02	2018	11	1	8
Farhan	Ali	2022-01-12	2026-10-02	2022	1	12	4
Gauri	Menon	2020-09-23	2026-10-02	2020	9	23	6
Harsh	Bansal	2021-04-18	2026-10-02	2021	4	18	5
Ishita	Das	2019-08-30	2026-10-02	2019	8	30	7
Jatin	Malhotra	2022-11-05	2026-10-02	2022	11	5	4
Lavanya	Pillai	2023-02-10	2026-10-02	2023	2	10	3

LAB 5 | SQL Operators

Arithmetic, Comparison and Logical Operators

QUERY

```
SELECT
    employee_id,
    first_name,
    department_id,

    employee_id + 10 AS id_plus_10,

    employee_id > 105 AS id_greater_than_105,

    department_id = 2 AND employee_id > 105 AS dept2_and_id_above_105,
    department_id = 2 OR department_id = 3 AS dept2_or_dept3,
    NOT department_id = 2 AS not_dept2

FROM employees;
```

OUTPUT

employee_id	first_name	department_id	id_plus_10	id_greater_than_105	dept2_and_id_above_105	dept2_or_dept3	not_dept2
101	Arjun	1	111	0	0	0	1
102	Bhavna	2	112	0	0	1	0
103	Chirag	3	113	0	0	1	1
104	Esha	4	114	0	0	0	1
105	Farhan	2	115	0	0	1	0
106	Gauri	3	116	1	0	1	1
107	Harsh	1	117	1	0	0	1
108	Ishita	4	118	1	0	0	1
109	Jatin	3	119	1	0	1	1
110	Lavanya	2	120	1	1	1	0

LAB 6 | Special Operators & Set Operations

Special Operators

QUERY

```sql
SELECT
    employee_id,
    first_name,
    department_id,
    email
FROM employees
WHERE employee_id BETWEEN 103 AND 108
   OR department_id IN (1, 3)
   OR first_name LIKE 'A%'
   OR department_id IS NULL;
```

OUTPUT

employee_id	first_name	department_id	email
101	Arjun	1	arjun.v@company.com
103	Chirag	3	chirag.d@company.com
104	Esha	4	esha.k@company.com
105	Farhan	2	farhan.a@company.com
106	Gauri	3	gauri.m@company.com
107	Harsh	1	harsh.b@company.com
108	Ishita	4	ishita.d@company.com
109	Jatin	3	jatin.m@company.com

Set Operations - Union

QUERY

```sql
SELECT first_name
FROM employees
WHERE department_id = 1

UNION

SELECT first_name
FROM employees
WHERE department_id = 3;
```

OUTPUT

first_name
Arjun
Harsh
Chirag
Gauri
Jatin

Set Operations - Intersect

QUERY

```
SELECT department_id
FROM employees
WHERE employee_id <= 105

INTERSECT

SELECT department_id
FROM employees
WHERE employee_id >= 103;
```

OUTPUT

department_id
1
2
3
4